

Basics

in general:
 $u = u(x, y, z)$

Laplacian: $\Delta u = u_{xx} + u_{yy} + u_{zz}$

Gradient: $\nabla u = (u_x, u_y, u_z)^T$

Burger's eq.: $u_t - u u_x = 0$ Poisson eq.: $\Delta u = f(x)$

Wave eq.: $u_{tt} - c^2 \Delta u = 0$ Transport eq.: $u_t + c u_x = 0$

Heat eq.: $u_t - k \Delta u = 0$ Laplace eq.: $\Delta u = 0$

Well-posed problem: Coherently ill-posed

- 1) Existence: Problem has a solution
- 2) Uniqueness: Problem has only one solution
- 3) Stability: Small change in problem leads to small change in solution.

Strong solution: All derivatives in the POE exist and are continuous on the whole domain of the PDE

Weak solution: 3 points in the domain where the derivatives do not exist/are not continuous. Cannot be plugged into the PDE

Classification properties of PDEs

- Order: Order of the highest derivative
- Linearity: If all coefficients of u and its derivatives do not depend on u or its derivatives $\rightarrow$ allows superposition
- Quasi-linearity: Linearity in highest order derivative term
- Homogeneity: $f(x) = 0$

ODE solving methods

linear ODE w. constant coeff. $(a_n x^{(n)} + \dots + a_0 x = f)$
 $\rightarrow x = x_p + x_h \Rightarrow x_h = e^{\lambda t}$ (Ansatz), $x_h = A e^{\lambda t} + \dots$
 $\Rightarrow x_p \sim$ similar to f

separable ODE $(\frac{dy}{dx} = f(x)g(y))$

$G(y) = \int \frac{1}{g(y)} dy = \int f(x) dx = F(x)$ (solve for y)

Integrating factor $(x'(t) + a(t)x(t) = b(t))$

Let $A'(t) = a(t)$: $x(t) = e^{-A(t)} \int_0^t e^{A(r)} b(r) dr$ and with $x(0) = x_0$
 $+ e^{A(0)} - A(t) \cdot x_0$

Classification of 2nd order PDE

$L[u] := a u_{xx} + 2b u_{xy} + c u_{yy} + d u_x + e u_y + f u = g$

Examples: $\delta = b^2 - ac$

Hyperbolic: $\delta > 0$ (Wave eq.)

Parabolic: $\delta = 0$ (Heat eq.)

Elliptic: $\delta < 0$ (Laplace eq.)

Method of Characteristics

LD for 1st order (quasilinear) PDEs

$\{ a(x, y) u_x(x, y) + b(x, y) u_y(x, y) = c(x, y, u) \}$

$u(x_0, y_0) = \tilde{u}_0$

1) Initial curve: $\Gamma(s) = (x_0(s), y_0(s), \tilde{u}_0(s))$

2) Solve ODE system: $\equiv (x(s, 0), y(s, 0), \tilde{u}(s, 0))$

$\{ \begin{aligned} x_t(t, s) &= a(x, y) \\ y_t(t, s) &= b(x, y) \\ u_t(t, s) &= c(x, y, u) \end{aligned}$ solve for $t(x, y)$ and $s(x, y)$

4) $u(x, y) = \tilde{u}(t, s)$

Transversality condition:

$$\det \begin{bmatrix} x_t(0, s) & y_t(0, s) \\ x_s(0, s) & y_s(0, s) \end{bmatrix} = \det \begin{bmatrix} a_0(s) & b_0(s) \\ x'_0(s) & y'_0(s) \end{bmatrix} \neq 0$$

$\Rightarrow$ Gives local existence and uniqueness around $(x_0(s), y_0(s))$

Obstacles to global existence:

- 1) solutions can blow up in finite time
- 2) characteristics cross the initial curve
- 3) derivatives cross each other

Conservation laws:

Def: PDEs describing the evolution of conserved quantities. x is the spacial variable, t the time variable.

$\{ u_t + c(u) u_x = 0 \Leftrightarrow u_t + \partial_x f(u) = 0, (x, y) \in \mathbb{R}^n \times (0, \infty) \}$

$u(x, 0) = u_0(x)$

Flux function: $f(u): \mathbb{R} \rightarrow \mathbb{R}$, Velocity $c(u) = f'(u)$

• characteristics are straight lines

• Traveling waves: satisfy this equation:
 $\forall y \in [0, y_c) \quad u(x, y) = u_0(x - c(y) \cdot y)$

Burger's eq.: $c(u) = u, f(u) = \frac{1}{2} u^2$

Transport eq.: $c \in \mathbb{R}, f(u) = c \cdot u$

Critical time

The transversality condition guarantees local existence and uniqueness on $[0, y_c)$.

$$y_c = \begin{cases} -\inf_{s \in \mathbb{R}} \frac{1}{c'(u_0(s)) \cdot u_0'(s)} & \text{if } \partial_s c(u_0(s)) > 0 \forall s \in \mathbb{R} \\ \infty & \text{if } \partial_s c(u_0(s)) \leq 0 \forall s \in \mathbb{R} \end{cases}$$

• If $u(x, 0)$ never decreases $\rightarrow$ no critical time

• If $y_c < 0$, no critical time

LD after $y_c \Rightarrow$ weak solution

The notion of weak solutions

$$\int_a^b [u(x, y)]_{y=y_a}^{y=y_b} dx = - \int_{y_a}^{y_b} \int_a^b [f(u(x, y))]_{x=a}^{x=b} dy$$

$\rightarrow$ holds even if u isn't smooth on the whole domain D

• weak solutions satisfy the POE in each subdomain D_i of D ($D = \bigcup_{i=1}^n D_i$).

• boundaries between D_i are called shocks

Rankine-Hugoniot condition

$x = \eta(y)$ a smooth curve across which $u(x, y)$ is discontinuous. η is called shock wave.

$$\eta'(y) = \frac{f(u^+) - f(u^-)}{u^+ - u^-}, \quad u^\pm = \lim_{x \rightarrow \eta(y)^\pm} u(x, y)$$

(= speed of shock wave)

• u^+/u^- solution on the right/left side

General approach

1) find y_c 2) use MoC to find strong solution

3) for $y > y_c$: Find characteristic crossing (x_0, y_c)

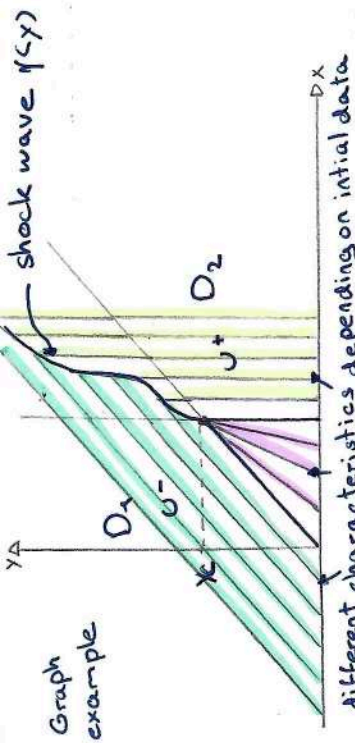
$\rightarrow$ solve: $x_0 = \eta(y) = \eta'(y_c) y_c + b$ (Find b)

$\rightarrow u(x, y) = \begin{cases} u^-, & x < \eta(y) \\ u^+, & x > \eta(y) \end{cases}$

4) use entropy condition if there is no unique weak solution:

$$c(u^+) < \eta'(y) < c(u^-) \Leftrightarrow f'(u^+) < \eta'(y) < f'(u^-)$$

($\rightarrow$ characteristics must enter the shock)



Hyperbolic PDEs: The Wave eq. in $\mathbb{R}$

General solution (even if F/G are not smooth):

$$u(x, t) = F(x+ct) + G(x-ct)$$

- G is called forward wave (if $c > 0$)
- F is called backward wave
- $\rightarrow$ are constant along the characteristics
- $x+ct = \alpha \in \mathbb{R}$ and $x-ct = \beta \in \mathbb{R}$

Let $g(x, t) = x+ct$, $h(x, t) = x-ct$ and

$$w(\xi, \eta) = u(x(\xi, \eta), y(\xi, \eta)). \text{ Recall: } w_t =$$

$$w_\xi \frac{\partial \xi}{\partial t} + w_\eta \frac{\partial \eta}{\partial t} \Rightarrow u_t = c^2(w_{\xi\xi} - 2w_{\xi\eta} + w_{\eta\eta})$$

and $u_{xx} = w_{\xi\xi} + 2w_{\xi\eta} + w_{\eta\eta} \Rightarrow u_t - c^2 u_{xx} = -4c^2 w_{\xi\eta} = 0$
 $w_{\xi\eta} = 0$ is the canonical form of the wave eq.

D'Alembert's formula

The solution to:
$$\begin{cases} u_{tt} - c^2 u_{xx} = F(x, t) & (x, t) \in \mathbb{R} \times \mathbb{R}^+ \\ u(x, 0) = f(x) & x \in \mathbb{R} \\ u_t(x, 0) = g(x) & x \in \mathbb{R} \end{cases}$$

is given by:

$$u(x, t) = \frac{f(x+ct) + f(x-ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(y) dy + \frac{1}{2c} \int_0^t \int_{x-c(t-\tau)}^{x+c(t-\tau)} F(\xi, \tau) d\xi d\tau$$

Simple inhomogeneities ($F(x)$ or $F(t) = F(x, t)$)

$\rightarrow$ find $v(x)/v(t)$ with $F(x) = -c^2 v_{xx}/F(t) = v_{tt}$

$\rightarrow$ use superposition: $u(x, t) = v(x, t) + w(x, t)$

Theorems:

- Uniqueness: The 1D-wave eq. has a unique sol.
- Singularities propagate along characteristics
- If g and f are even/odd/periodic $\Rightarrow u$ is also even/odd/periodic (possibility to extend $f(x)$ and $g(x)$ oddly or evenly)

The domain of dependence

Given by $[x_0-ct_0, x_0+ct_0]$

for a point (x_0, t_0)

$\rightarrow$ value changes if the values in the interval change

The region of influence

All points satisfying $x-ct \leq b$, $x+ct \geq a$ are dependent on the initial condition in $[a, b]$

Method of Separation of Variables

Ansatz always the same: Insert $u(x, t) = X(x)T(t)$ into the PDE and decouple it into two ODEs.

$X''(x) = -\lambda X(x)$ is recurring. Solution are:

$$X(x) = \begin{cases} A \sin(\sqrt{\lambda} x) + B \cos(\sqrt{\lambda} x), & \lambda > 0 \\ C + Dx, & \lambda = 0 \\ A \sinh(\sqrt{-\lambda} x) + B \cosh(\sqrt{-\lambda} x), & \lambda < 0 \end{cases}$$

Boundary conditions used to find the non-trivial solutions:

Dirichlet: $u(0, t) = u(L, t) = 0$

Neumann: $u_x(0, t) = u_x(L, t) = 0$

Mixed 1: $u(0, t) = u_x(L, t) = 0$

Mixed 2: $u_x(0, t) = u(L, t) = 0$

$X_n(x) = \sin(\sqrt{\lambda_n} x)$

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

$$X_n(x) = \cos(\sqrt{\lambda_n} x)$$

$$\lambda_n = \left(\frac{n+\frac{1}{2}\pi}{L}\right)^2$$

$$\lambda_n = \left(\frac{n+\frac{1}{2}\pi}{L}\right)^2$$

Homogeneous 1D heat equation

Let $K \in \mathbb{R}^+$ be the constant of diffusivity:

$$\begin{cases} u_t - Ku_{xx} = 0, & (x, t) \in (0, L) \times (0, \infty) \\ \text{Boundary condition, } t \in \mathbb{R}^+ \\ u(x, 0) = f(x), & x \in (0, L) \end{cases}$$

Separation of variable gives:

$$\begin{cases} X'' = -\lambda X, & x \in (0, L) \\ T' = -\lambda K T, & t > 0 \end{cases} \Rightarrow T_n(t) = A_n e^{-\lambda_n K t}$$

General solution:

$$\begin{aligned} \text{Dirichlet \& Mixed 1: } u(x, t) &= \sum_{n=1}^{\infty} X_n(x) T_n(t) \\ \text{Neumann \& Mixed 2: } u(x, t) &= A_0 + \sum_{n=1}^{\infty} X_n(x) T_n(t) \end{aligned}$$

2) Apply initial condition $u(x, 0)$:

$$A_n = \frac{2}{L} \int_0^L f(x) \dots dx$$

Homogeneous 1D wave equation

$$u_{tt} - c^2 u_{xx} = 0, \quad (x, t) \in [0, L] \times [0, \infty)$$

Boundary condition, $t > 0$

$$u(x, 0) = f(x), \quad x \in \mathbb{R}$$

$$u_t(x, 0) = g(x), \quad x \in \mathbb{R}$$

Separation of variable gives:

$$X'' = -\lambda X, \quad x \in (0, L)$$

$$T'' = -\lambda c^2 T, \quad t > 0$$

$$\Rightarrow T_n = A_n \cos(ct\sqrt{\lambda_n}) + B_n \sin(ct\sqrt{\lambda_n})$$

1) general solution:

$$\text{Dirichlet \& Mixed 1: } u(x, t) = \sum_{n=1}^{\infty} X_n(x) T_n(t)$$

$$\text{Neumann \& Mixed 2: } u(x, t) = A_0 + B_0 t + \sum_{n=1}^{\infty} X_n(t) T_n(t)$$

2) Apply initial condition $u(x, 0)$ and $u_t(x, 0)$:

$$A_n = \frac{2}{L} \int_0^L f(x) \dots dx \quad B_n = \frac{2}{cn\pi} \int_0^L g(x) \dots dx$$

$$(B_0 = \frac{2}{L} \int_0^L g(x) dx)$$

Inhomogeneous boundary condition

Find a $w(x, t)$ that satisfies the boundary condition.

Define $v = w + u$

Solve for v , combine solutions

Inhomogeneous heat/wave eq.

- 1) Use solution for the homogeneous case without solving for T
- 2) Express the inhomogeneity in the corresponding basis:

ex: $\sum_n T_n'' X_n - c^2 \sum_n T_n X_n = h(x, t) = \sum_n c_n(t) X_n$

$$\sum_n T_n(0) X_n = f(x) = \sum_n a_n X_n$$

$$\sum_n T_n'(0) X_n = g(x) = \sum_n b_n X_n$$

$$\Rightarrow \begin{cases} T_n'' + c^2 \left(\frac{n\pi}{L}\right)^2 T_n = c_n \\ T_n(0) = \frac{2}{L} \int_0^L X_n f(x) dx \\ T_n'(0) = \frac{2}{L} \int_0^L X_n g(x) dx \end{cases}$$

Uniqueness with energy method

Let u_1 and u_2 be solution to the PDE

• Energy function: $E(t) = \int_0^L \omega_1^2 + c^2 \omega_2^2 dx$ (for wave eq.)

• Show that $\frac{d}{dt} E(t) = 0$ and $E(0) = 0$

$\Rightarrow w = 0 \Rightarrow u_1 = u_2$ (solution is unique)

Elliptic PDEs - Laplace & Poisson eq.

$u(x, y)$ is called harmonic if it solves the Laplace eq.

Dirichlet: $\begin{cases} \Delta u = p(x, y) & (x, y) \in D \\ u = g(x, y) & (x, y) \in \partial D \end{cases}$

Neumann: $\begin{cases} \Delta u = p(x, y) & (x, y) \in D \\ \partial_n u = \vec{\nu} \cdot \nabla u = g(x, y) & (x, y) \in \partial D \end{cases}$

Mixed: $\begin{cases} \Delta u = p(x, y) & (x, y) \in D \\ u = g(x, y) & (x, y) \in \partial D \end{cases}$

Compatibility condition for Neumann: $\int_{\partial D} \partial_n u = \int_{\partial D} g(x, y) ds = 0$

necessary for existence of a solution:

$$\int_0^{2\pi} \partial_n u = \int_0^{2\pi} g(\theta) d\theta = 0$$

for circles:

$$\int_{\partial D} \partial_n u = \int_0^{2\pi} \partial_n u(r, \theta) r d\theta = \int_0^{2\pi} g(r, \theta) r d\theta = 0$$

Maximum Principles

Let $x_0 \in D$ be a maximum point, then:

$$\nabla u(x_0) = 0 \text{ (gradient)} \text{ and } D^2 u(x_0) \leq 0 \text{ (Hessian)}$$

Weak maximum principle

Let D be a bounded domain and let $u(x, y) \in C^2(D) \cap C(\bar{D})$ be a harmonic function in D . The maximum (and minimum) of u in $\bar{D}$ is achieved on ∂D :

$$\max_{\bar{D}} u = \max_{\partial D} u \quad \min_{\bar{D}} u = \min_{\partial D} u$$

Mean value principle

Let u be a harmonic function on D and let $B_R(x_0, y_0) \subset D$ be a ball of radius R . $\Leftrightarrow$

$$u(x_0, y_0) = \frac{1}{2\pi} \int_0^{2\pi} u(x_0 + R \cos \theta, y_0 + R \sin \theta) d\theta$$

The strong maximum principle

Let u be a harmonic in D and D a connected subset of $\mathbb{R}^2$, the Dirichlet problem

$$\begin{cases} \Delta u = f & \text{in } D \\ u = g & \text{on } \partial D \end{cases} \text{ has at most one solution}$$

Theorem: Let $u_1, u_2 \in C^2(D) \cap C(\bar{D})$ solve

$$\begin{cases} \Delta u_1 = 0 \text{ in } D \\ u_1 = g_1 \text{ on } \partial D \end{cases} \quad \begin{cases} \Delta u_2 = 0 \text{ in } D \\ u_2 = g_2 \text{ on } \partial D \end{cases}$$

$$\Rightarrow \max_{\bar{D}} |u_1 - u_2| = \max_{\partial D} |g_1 - g_2|$$

Maximum principle for parabolic equations

$$0 = u_t - k u_{xx} \Leftrightarrow u_t = k \Delta u$$

problem lives in space time $Q_T = [0, T] \times D$ $x \in D, t \in [0, T]$. Define the parabolic boundary

$$\partial_p Q_T = \{0\} \times D \cup \{0, T\} \times \partial D$$

$\Rightarrow u$ achieves its maximum and minimum on $\partial_p Q_T$

Laplace eq. in rectangular and circular domains

Combinations of Dirichlet and Neumann also possible. $\rightarrow$ choose the right bases!

Dirichlet problem with rectangular domain

$$\begin{cases} \Delta u = 0 & (x, y) \in (a, b) \times (c, d) \\ u(a, y) = f(y) & (x, y) \in \{a\} \times [c, d] \\ u(b, y) = g(y) & (x, y) \in \{b\} \times [c, d] \\ u(x, c) = h(x) & (x, y) \in [a, b] \times \{c\} \\ u(x, d) = k(x) & (x, y) \in [a, b] \times \{d\} \end{cases}$$

$\rightarrow$ split into two subproblems $u = u_1 + u_2$

$$\begin{cases} \Delta u_1 = 0 \\ u_1(a, y) = f(y) \\ u_1(b, y) = g(y) \\ u_1(x, c) = 0 \\ u_1(x, d) = 0 \end{cases} \quad \begin{cases} \Delta u_2 = 0 \\ u_2(a, y) = 0 \\ u_2(b, y) = 0 \\ u_2(x, c) = h(x) \\ u_2(x, d) = k(x) \end{cases}$$

Separation of variables (switch x & y for u_2)

$$X'' - \lambda X = 0 \quad \begin{cases} Y'' + \lambda Y = 0 \\ Y(c) = Y(d) = 0 \end{cases}$$

1) General solutions:

$$u_1 = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{d-c}(y-c)\right) \left[A_n \sinh\left(\frac{n\pi}{d-c}(x-a)\right) + B_n \sinh\left(\frac{n\pi}{d-c}(x-b)\right) \right]$$

$$u_2 = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{b-a}(x-a)\right) \left[C_n \sinh\left(\frac{n\pi}{b-a}(y-c)\right) + D_n \sinh\left(\frac{n\pi}{b-a}(y-d)\right) \right]$$

2) Use initial conditions to find coefficients.

Issue for numerical approach

By splitting the problem the boundary data may not be continuous anymore. The trick to avoid this issue is defining

$\bar{u} = u - p$ where $p(x, y) = a_0 + a_1 x + a_2 y + a_3 xy$ is harmonic. $\rightarrow$ choose a_i so that at the corners $\bar{u} = 0$.

Neumann problem with rectangular domain

$u = C \in \mathbb{R}$ is always a solution $\rightarrow \infty$ solutions due to superposition.

$$\begin{cases} \Delta u = 0 & (x, y) \in (a, b) \times (c, d) \\ u_x(a, y) = f(y) & (x, y) \in \{a\} \times [c, d] \\ u_x(b, y) = g(y) & (x, y) \in \{b\} \times [c, d] \\ u_y(x, c) = h(x) & (x, y) \in [a, b] \times \{c\} \\ u_y(x, d) = k(x) & (x, y) \in [a, b] \times \{d\} \end{cases}$$

compatibility condition:

$$\int_a^b [u_x(x, y)]_{y=c}^{y=d} dy + \int_c^d [u_y(x, y)]_{x=a}^{x=b} dx = 0$$

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Once again we want to split the problem into subproblems. But this can lead to problems in the compatibility condition. Therefore, we add a harmonic polynomial:
 $\rightarrow v = u + \alpha(x^2 - y^2) \rightarrow v = v_1 + v_2$

$$\begin{cases} \Delta v_1 = 0 \\ (v_1)_x(\alpha, \gamma) = f(\gamma) + 2\alpha\alpha \\ (v_1)_x(b, \gamma) = g(\gamma) + 2\alpha b \\ (v_1)_y(x, c) = 0 \\ (v_1)_y(x, d) = 0 \end{cases} \quad \Delta v_2 = 0 \quad \begin{cases} (v_2)_x(\alpha, \gamma) = 0 \\ (v_2)_x(b, \gamma) = 0 \\ (v_2)_y(x, c) = k(x) - 2\alpha c \\ (v_2)_y(x, d) = h(x) - 2\alpha d \end{cases}$$

using the compatibility condition to find α :

$$\int_c^d g + 2\alpha b - \int_c^d f + 2\alpha a = 0, \alpha = \frac{1}{2(b-a)(d-c)} \int_c^d f - g$$

1) general solution given by:

$$v_1 = A_0 x + B_0 + \sum_{n=1}^{\infty} \cos\left(\frac{n\pi}{d-c}(y-c)\right) \cdot \left[A_n \cosh\left(\frac{n\pi}{d-c}(x-a)\right) + B_n \cosh\left(\frac{n\pi}{d-c}(x-b)\right) \right]$$

$$v_2 = C_0 y + D_0 + \sum_{n=1}^{\infty} \cos\left(\frac{n\pi}{b-a}(x-a)\right) \cdot \left[C_n \cosh\left(\frac{n\pi}{b-a}(x-c)\right) + D_n \cosh\left(\frac{n\pi}{b-a}(x-d)\right) \right]$$

2) use initial conditions to find coeff.

Laplace eq. in circular domains
 $\rightarrow$ for D with radial symmetry

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad A_0 = \int_0^{2\pi} \int_0^r w(r, \theta) \cdot r \, d\theta \, dr$$

where $r^2 = x^2 + y^2$ and $\sin^2 \theta + \cos^2 \theta = 1$

$$\Rightarrow \Delta u = w_r + \frac{1}{r} w_r + \frac{1}{r^2} w_{\theta\theta}$$

Approach for disk and annulus

Let the domain be $D := \{r \in [a, b], \theta \in [0, 2\pi]\}$
 With separation of variable $w(r, \theta) = R(r) \Theta(\theta)$

$$\begin{cases} r^2 R'' + r R' = \lambda R \\ \Theta'' = -\lambda \Theta \end{cases} \quad \begin{cases} \Theta(0) = \Theta(2\pi) \\ \Theta'(0) = \Theta'(2\pi) \end{cases}$$

The solutions are given by:

$$\Theta_n(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta), \quad n \in \mathbb{N}$$

$$R_n(r) = \begin{cases} C_0 + D_0 \log r & n=0 \\ C_n r^n + D_n r^{-n} & n \neq 0 \end{cases}$$

1) general solution is:

$$w = A_0 + B_0 \log r + \sum_{n=1}^{\infty} r^n (A_n \cos(n\theta) + B_n \sin(n\theta)) + \sum_{n=1}^{\infty} r^{-n} (C_n \cos(n\theta) + D_n \sin(n\theta))$$

$\Rightarrow$ disregard r^{-n} and $\log r$ for disk, as they have a singularity at $r=0$

2) use boundary condition to find coefficients

Approach for disk sectors or annulus sector

Let the domain $D = \{r \in [a, b], \theta \in [0, \eta]\}$ with $\eta \in (0, 2\pi)$. We assume $w(r, 0) = w(r, \eta) = 0 \quad \forall r \in (a, b)$

Same operation but $\Theta(0) = \Theta(\eta) = 0$:

$$\Theta_n(\theta) = A_n \sin\left(\frac{n\pi}{\eta} \theta\right) \quad \lambda_n = \left(\frac{n\pi}{\eta}\right)^2$$

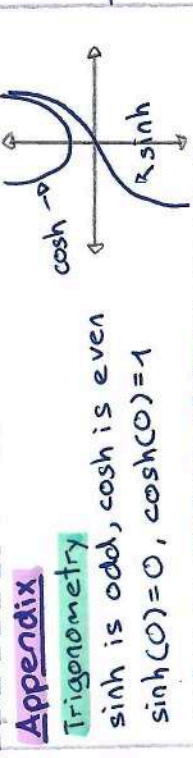
$$R_n(r) = C_n r^{n\pi/\eta} + D_n r^{-n\pi/\eta}$$

1) General solution becomes

$$w = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{\eta} \theta\right) r^{n\pi/\eta} + B_n \sin\left(\frac{n\pi}{\eta} \theta\right) r^{-n\pi/\eta}$$

2) use boundary conditions for coeff.

Appendix
 Trigonometry



$$\begin{aligned} \cos(x \pm y) &= \cos x \cos y \mp \sin x \sin y \\ \sin(x \pm y) &= \sin x \cos y \pm \sin y \cos x \\ \cos\left(x + \frac{\pi}{2}\right) &= -\sin x \quad \sin\left(x + \frac{\pi}{2}\right) = \cos x \\ \sin(x) \sin y &= \frac{1}{2} (\cos(x-y) - \cos(x+y)) \\ \cos(x) \cos y &= \frac{1}{2} (\cos(x-y) + \cos(x+y)) \\ \sin(x) \cos y &= \frac{1}{2} (\sin(x-y) + \sin(x+y)) \\ \sin^2(x) &= \frac{1}{2} (1 - \cos(2x)) \quad \cos^2(x) = \frac{1}{2} (1 + \cos(2x)) \end{aligned}$$

$$\begin{aligned} \sin^3(x) &= \frac{1}{4} (3 \sin x - \sin(3x)) \\ \cos^3(x) &= \frac{1}{4} (3 \cos x + \cos(3x)) \\ \sinh(z) &= \frac{1}{2} (e^z - e^{-z}) \quad \cosh(z) = \frac{e^z + e^{-z}}{2} \\ \cos(z) &= \frac{1}{2} (e^{iz} + e^{-iz}) \end{aligned}$$

sin	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
cos	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0	$-1/2$	$-\sqrt{2}/2$	$-\sqrt{3}/2$	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞	$-1/\sqrt{3}$	-1	$-\sqrt{3}$	0

Analysis 2 stuff:

Min/Max von f auf dem Rand mit Parametrisierung

$\eta: [a, b] \rightarrow \mathbb{R}^n$ erfüllen $\frac{d}{dt}(f \circ \eta)(t) = 0$

Richtungsableitung:

$$\partial_v f(\vec{x}) = \lim_{h \rightarrow 0} \frac{f(\vec{x} + h\vec{v}) - f(\vec{x})}{h}$$

Wegintegral über Skalarfeld:

$$\int_{\gamma} f(x) dx = \int_a^b f(\gamma(t)) \|\dot{\gamma}(t)\| dt$$

Koordinatentransformation:

$$\int_{\mathbb{R}^2} f(x, y) dx dy = \int_{\Omega} f(\Phi(x)) |\det d\Phi| dx$$

Polar: $r, d\theta$ Kugel: $r^2 \sin \theta, dr d\theta d\phi$

Kettenregel: 1D: $(f \circ \gamma(s))_s = \partial_s f(\gamma(s))$

2D: $\frac{\partial}{\partial s} f(x(s), y(s)) = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s}$

$\frac{\partial^2}{\partial s^2} f = \frac{\partial^2 f}{\partial x^2} \frac{\partial x^2}{\partial s^2} + 2 \frac{\partial^2 f}{\partial x \partial y} \frac{\partial x}{\partial s} \frac{\partial y}{\partial s} + \frac{\partial^2 f}{\partial y^2} \frac{\partial y^2}{\partial s^2}$ (only for smooth functions)